

Assignment 01

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Section: 3

Ans to the Q No-1

Given regular expression $(a|b)^*cb\#$, using Direct method we have to convert it into DFA

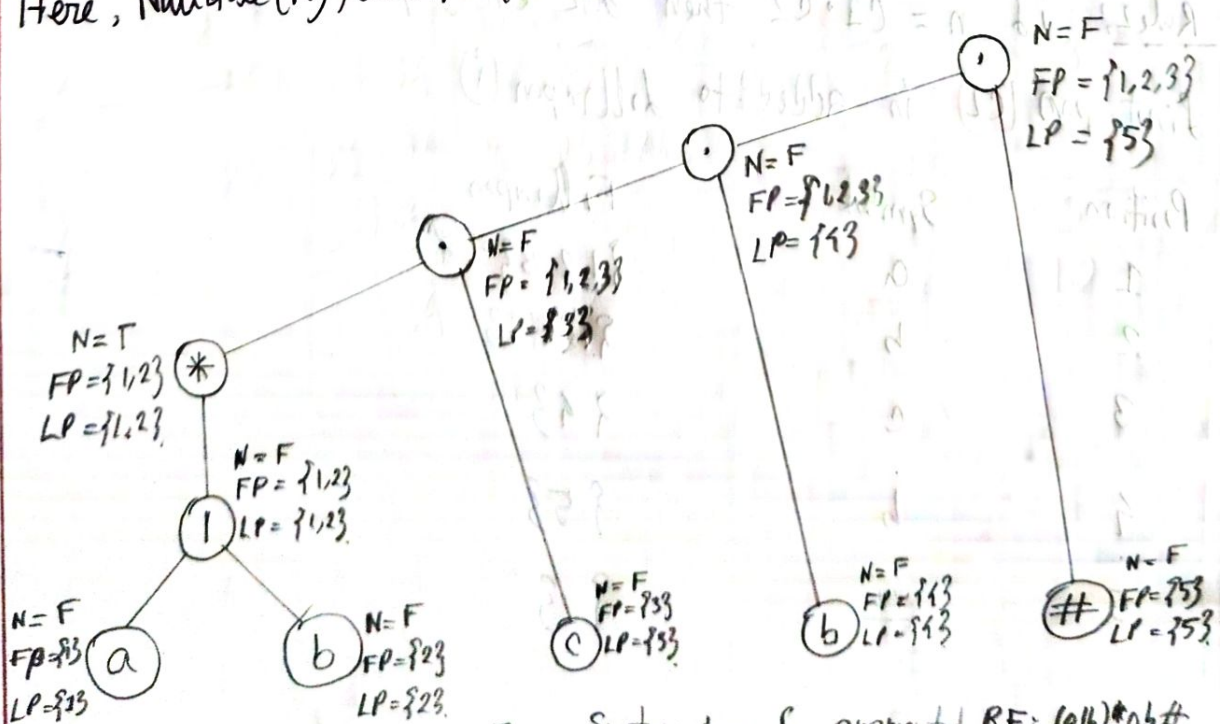
Step 1 - Augmentation

The regular expression is augmented with a unique end-marker $\#$, which is concatenated at the end. This guarantees that every accepting state of DFA corresponds to a position which includes the end of a valid string.

so, $(a|b)^*cb\#$

Step 2 - Syntax Tree, Nullable, Firstpos, Lastpos

Here, Nullable (N), Firstpos (FP) and Lastpos (LP) are computed bottom up for every node.



Summary table

Node	Nullable	First pos	Last pos
a(1)	F	{1}	{1}
b(2)	F	{2}	{2}
(a b)	F	{1,2}	{1,2}
(a b)*	T	{1,2}	{1,2}
c(3)	F	{3}	{3}
(a b)*c	F	{1,2,3}	{3}
b(4)	F	{4}	{4}
(a b)*cb	F	{1,2,3}	{4}
#(5)	F	{5}	{5}
Root: (a b)*cb#	F	{1,2,5}	{5}

Step 3- Followpos

Followpos is computed using two rules:

Rule 1: If n is a start node then for every position i in $\text{lastpos}(n)$,

$\text{firstpos}(n)$ is added to $\text{followpos}(i)$.

Rule 2: if $n = c_1 \cdot c_2$ then for every position i in $\text{lastpos}(c_1)$,

$\text{firstpos}(c_2)$ is added to $\text{followpos}(i)$

Position	Symbol	Followpos
1	a	{1,2,3}
2	b	{1,2,3}
3	c	{4}
4	b	{5}
5	#	{ }

3

Step 4 :- DFA construction

State transition table-

State	Position set	a	b	c
A (start)	{1,2,3}	A	A	B
B	{4}	Dead	C	Dead
C	{5}	Dead	Dead	Dead
Dead	-	Dead	Dead	Dead

Step 5 - DFA diagram

States are - A (start state), B, C (accepting state) and Dead (trap state)

